

# Machine Learning



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Haq

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# What is Machine Learning?

“Learning is any process by which a system improves performance from experience.”

- Herbert Simon

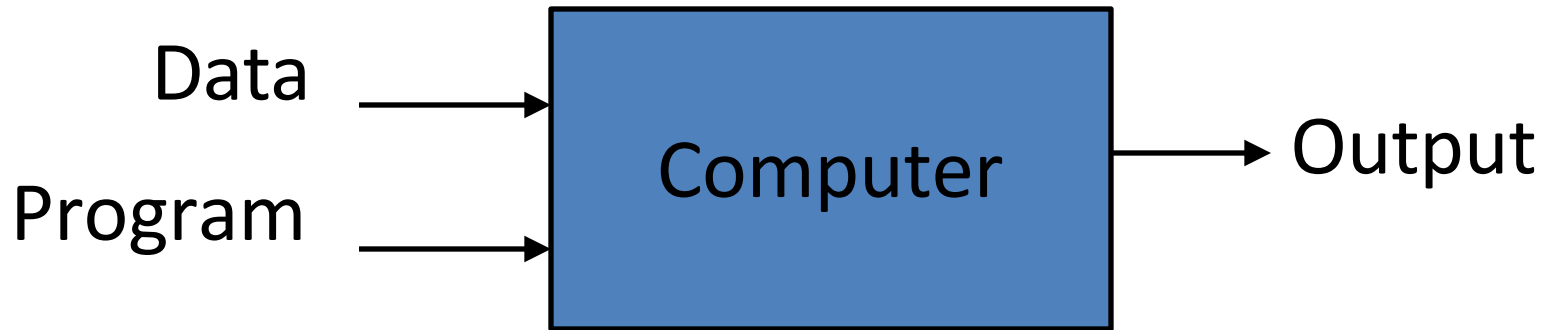
Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that

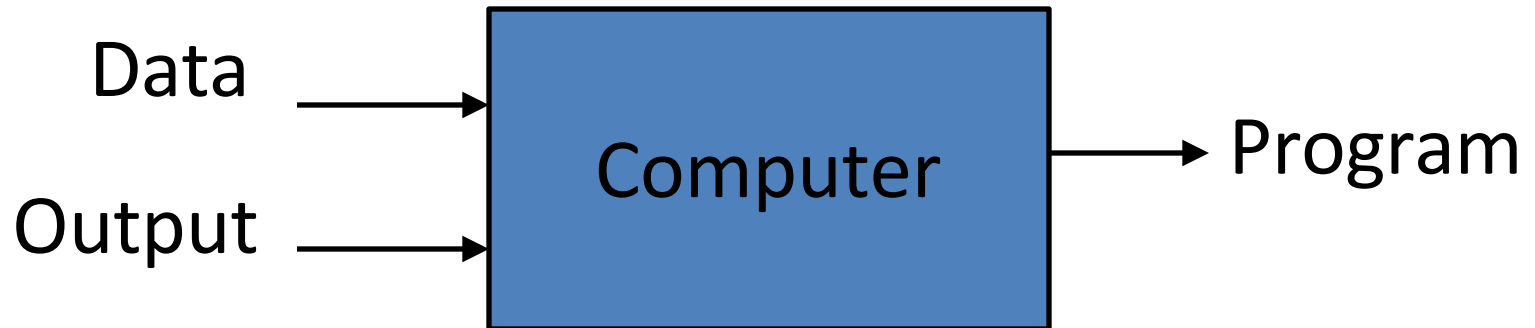
- improve their performance  $P$
- at some task  $T$
- with experience  $E$ .

A well-defined learning task is given by  $\langle P, T, E \rangle$ .

## Traditional Programming



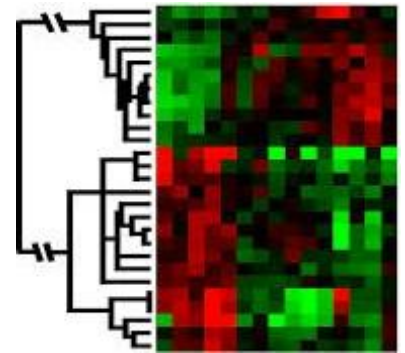
## Machine Learning



# When Do We Use Machine Learning?

ML is used when:

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)



Learning isn't always useful:

- There is no need to “learn” to calculate payroll

A classic example of a task that requires machine learning:  
It is very hard to say what makes a 2

0 0 0 1 1 1 1 1 1 2

2 2 2 2 2 2 2 3 3 3

3 4 4 4 4 4 5 5 5 5

6 6 7 7 7 7 8 8 8 8

9 9 9 9 9 9 9 9 9 9

# Some more examples of tasks that are best solved by using a learning algorithm

- Recognizing patterns:
  - Facial identities or facial expressions
  - Handwritten or spoken words
  - Medical images
- Generating patterns:
  - Generating images or motion sequences
- Recognizing anomalies:
  - Unusual credit card transactions
  - Unusual patterns of sensor readings in a nuclear power plant
- Prediction:
  - Future stock prices or currency exchange rates

# Sample Applications

- Web search
- Computational biology
- Finance
- E-commerce
- Space exploration
- Robotics
- Information extraction
- Social networks
- Debugging software
- [Your favorite area]

# Samuel's Checkers-Player

“Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.” -Arthur Samuel (1959)



# Defining the Learning Task

Improve on task T, with respect to  
performance metric P, based on experience E

T: Playing checkers

P: Percentage of games won against an arbitrary opponent

E: Playing practice games against itself

T: Recognizing hand-written words

P: Percentage of words correctly classified

E: Database of human-labeled images of handwritten words

T: Driving on four-lane highways using vision sensors

P: Average distance traveled before a human-judged error

E: A sequence of images and steering commands recorded while observing a human driver.

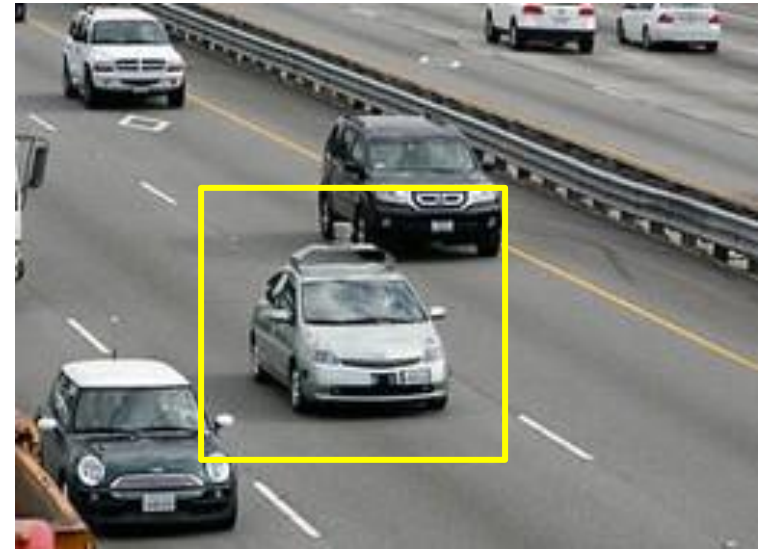
T: Categorize email messages as spam or legitimate.

P: Percentage of email messages correctly classified.

E: Database of emails, some with human-given labels

# State of the Art Applications of Machine Learning

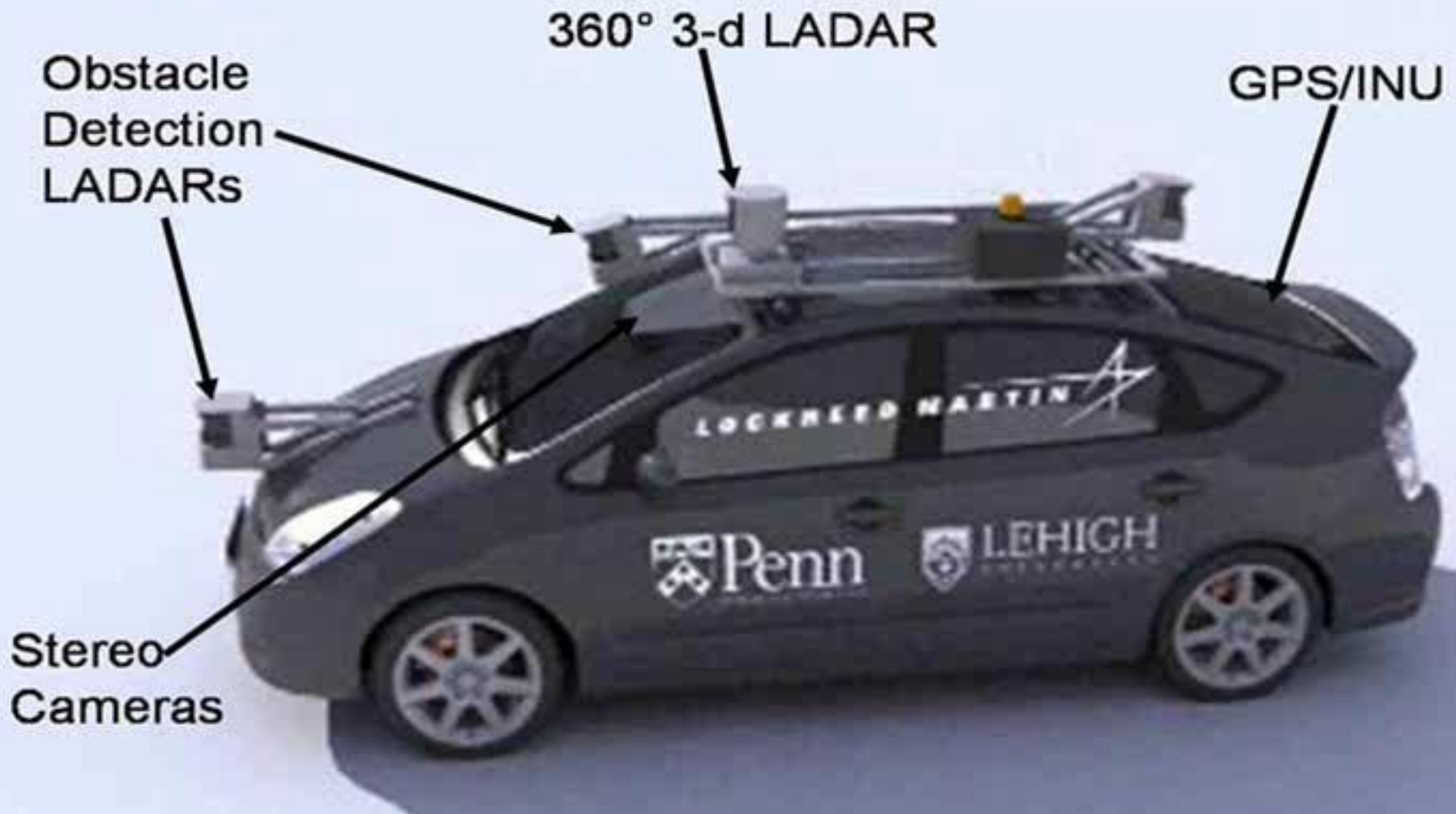
# Autonomous Cars



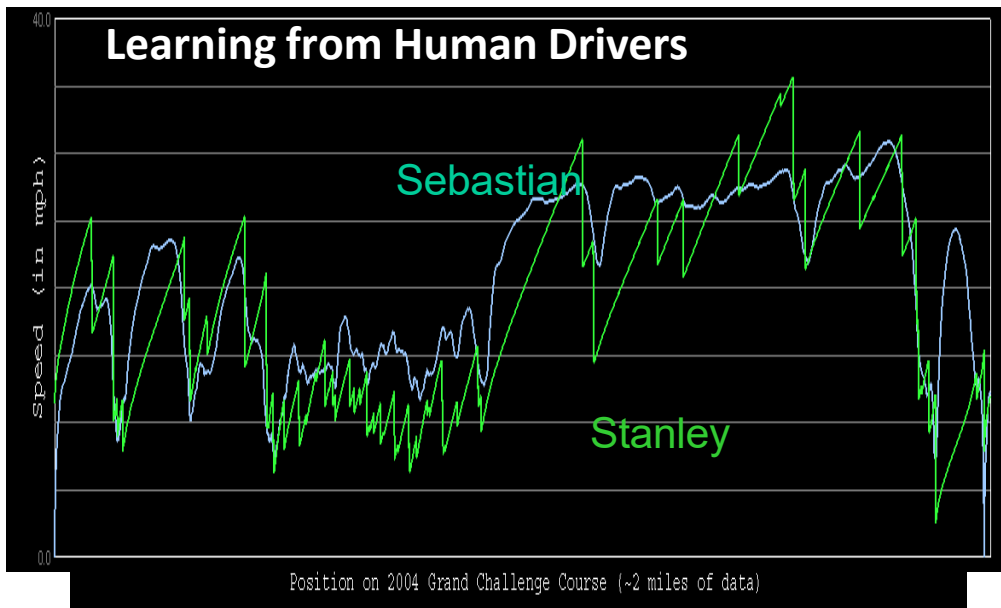
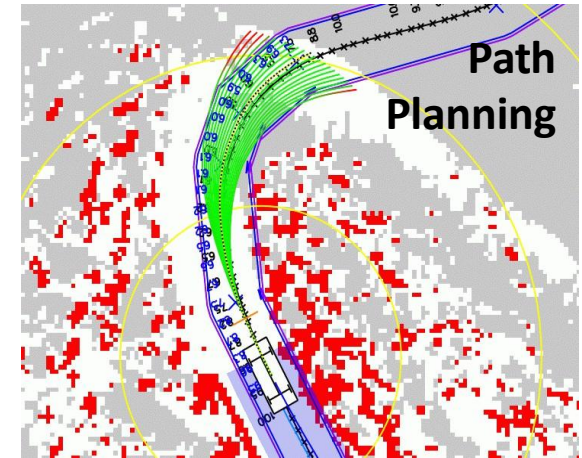
- Taiwan officially allowed self-driving cars to be tested on public roads in **2019**



# Autonomous Car Sensors



# Autonomous Car Technology



# Deep Learning in the Headlines

BUSINESS NEWS

MIT  
Technology  
Review

## Is Google Cornering the Market on Deep Learning?

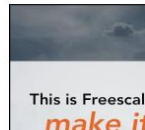
A cutting-edge corner of science is being wooed by Silicon Valley, to the dismay of some academics.

By Antonio Regalado on January 29, 2014



How much are a dozen deep-learning researchers worth? Apparently, more than \$400 million.

This week, Google [reportedly paid that much](#) to acquire [DeepMind Technologies](#), a startup based in



This is Freescale  
make it

## BloombergBusinessweek Technology

Acquisitions

## The Race to Buy the Human Brains Behind Deep Learning Machines

By Ashlee Vance | January 27, 2014

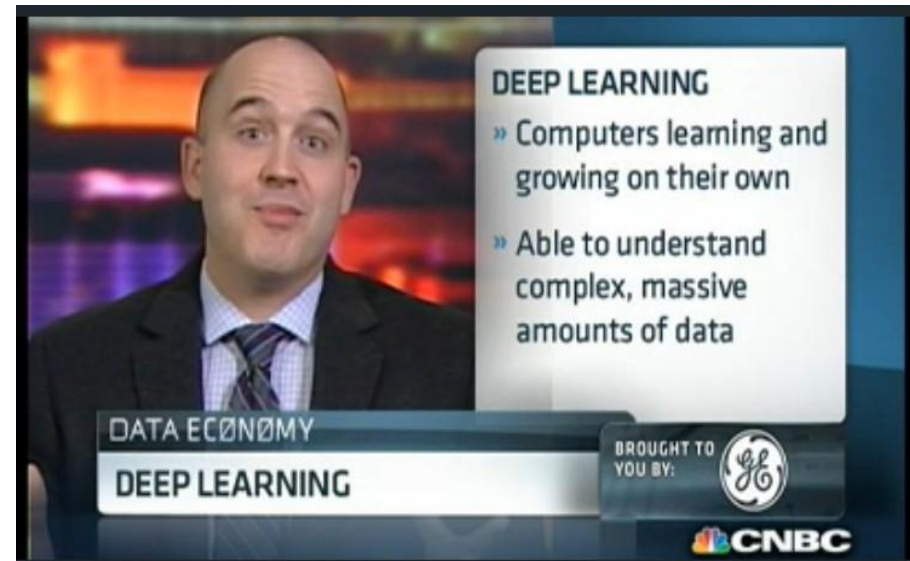
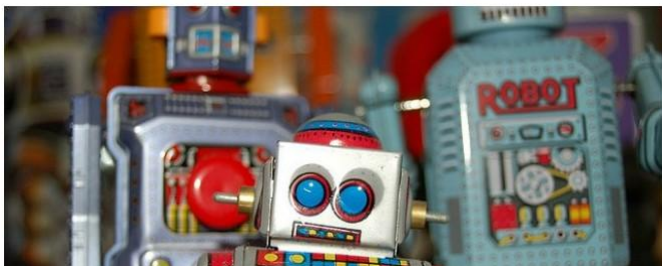
intelligence projects. “DeepMind is bona fide in terms of its research capabilities and depth,” says Peter Lee, who heads Microsoft Research.

According to Lee, Microsoft, Facebook ([FB](#)), and Google find themselves in a battle for deep learning talent. Microsoft has gone from four full-time deep learning experts to 70 in the past three years. “We would have more if the talent was there to

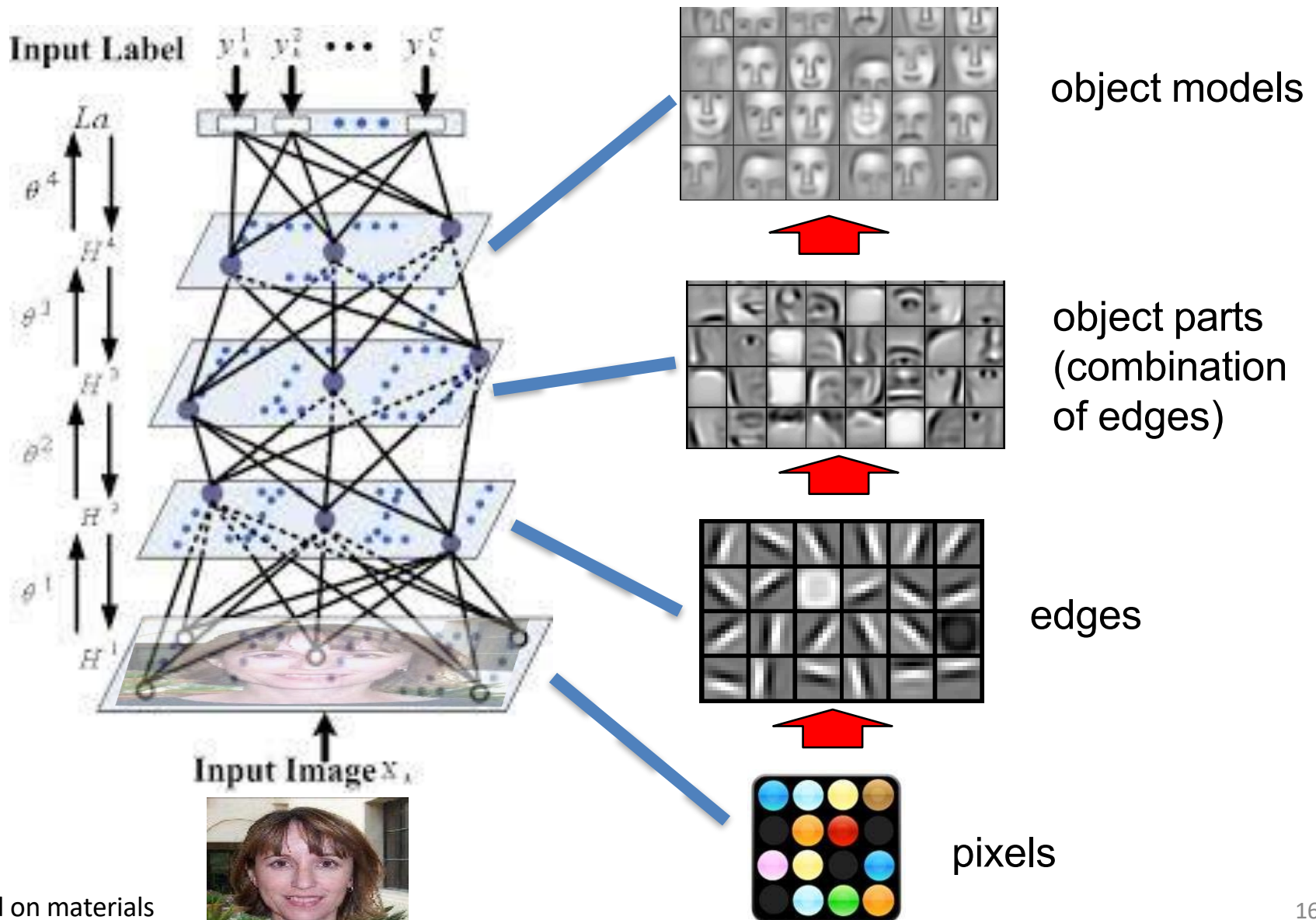


## Deep Learning's Role in the Age of Robots

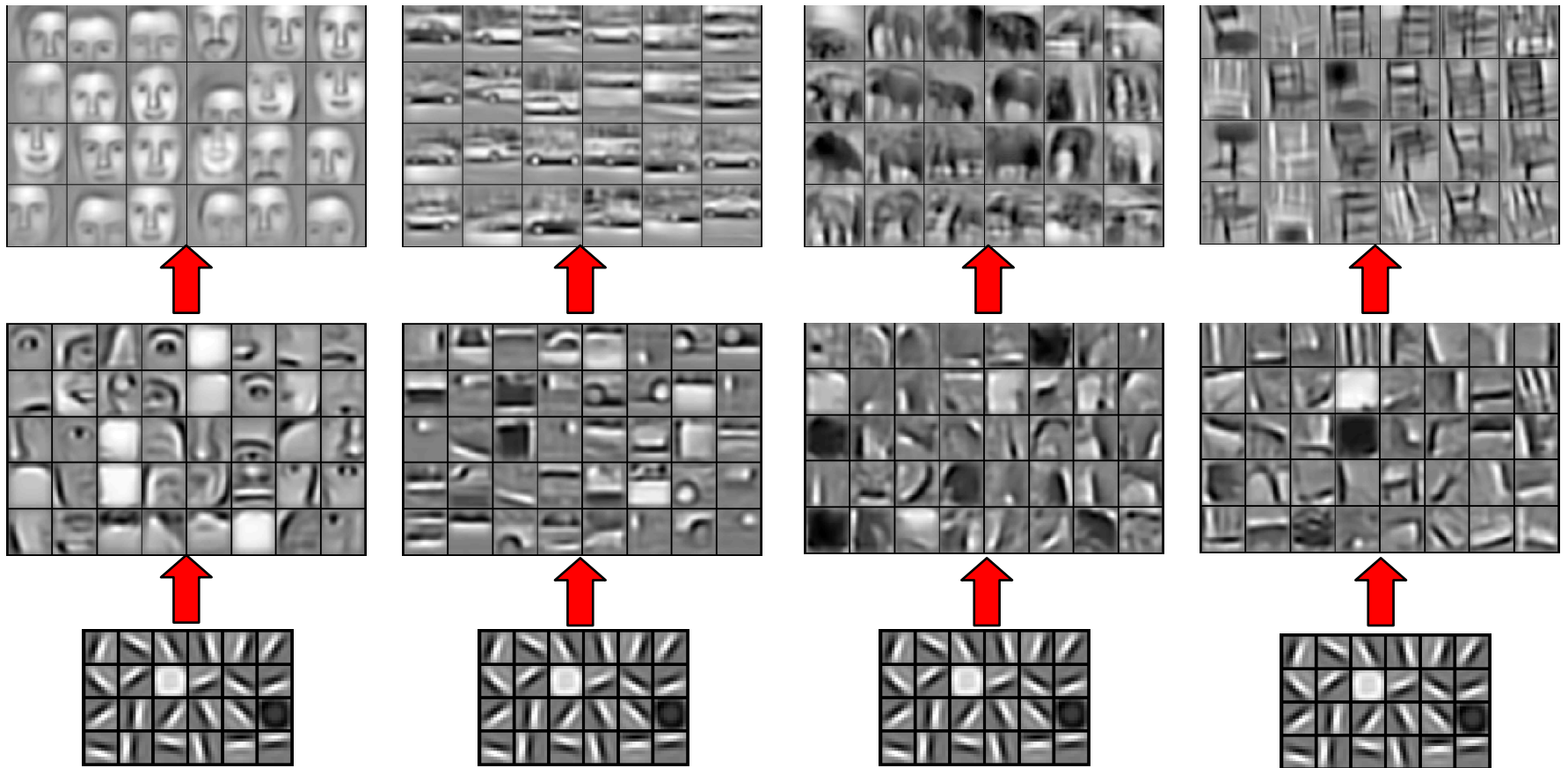
BY JULIAN GREEN, JETPAC 05.02.14 2:56 PM



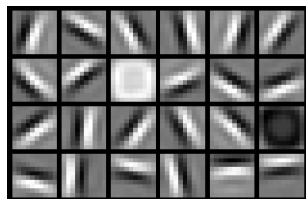
# Deep Belief Net on Face Images



# Learning of Object Parts



# Training on Multiple Objects



Trained on 4 classes (cars, faces, motorbikes, airplanes).

Second layer: Shared-features and object-specific features.

Third layer: More specific features.

# Scene Labeling via Deep Learning



# Inference from Deep Learned Models

Generating posterior samples from faces by “filling in” experiments (cf. Lee and Mumford, 2003). Combine bottom-up and top-down inference.

Input images



Samples from  
feedforward  
Inference  
(control)

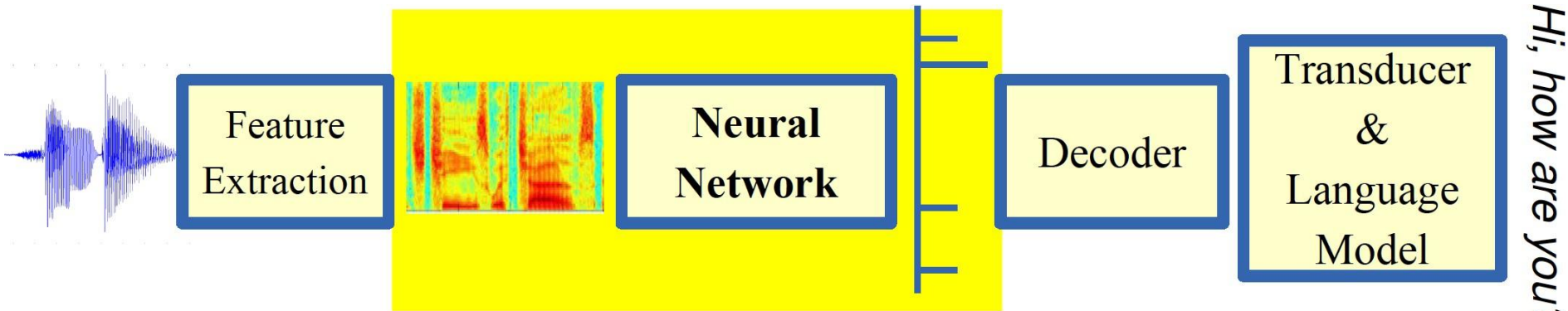


Samples from  
Full posterior  
inference

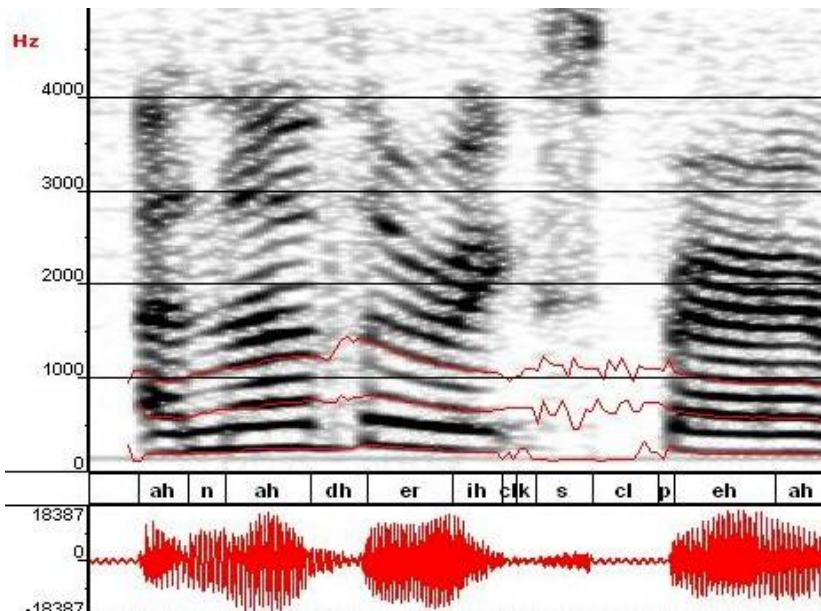


# Machine Learning in Automatic Speech Recognition

## A Typical Speech Recognition System



ML used to predict of phone states from the sound spectrogram



Deep learning has state-of-the-art results

| # Hidden Layers   | 1    | 2    | 4    | 8    | 10   | 12   |
|-------------------|------|------|------|------|------|------|
| Word Error Rate % | 16.0 | 12.8 | 11.4 | 10.9 | 11.0 | 11.1 |

Baseline GMM performance = 15.4%

[Zeiler et al. "On rectified linear units for speech recognition" ICASSP 2013]

# Impact of Deep Learning in Speech Technology



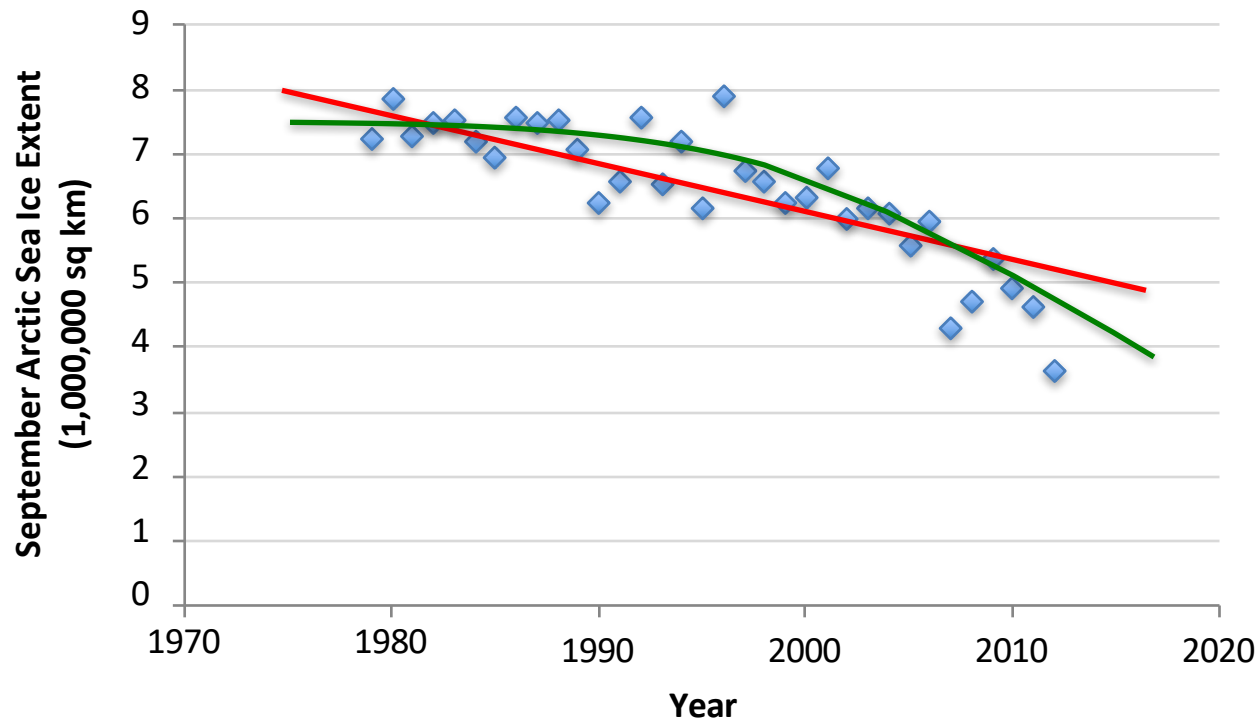
# Types of Learning

# Types of Learning

- **Supervised (inductive) learning**
  - Given: training data + desired outputs (labels)
- **Unsupervised learning**
  - Given: training data (without desired outputs)
- **Semi-supervised learning**
  - Given: training data + a few desired outputs
- **Reinforcement learning**
  - Rewards from sequence of actions

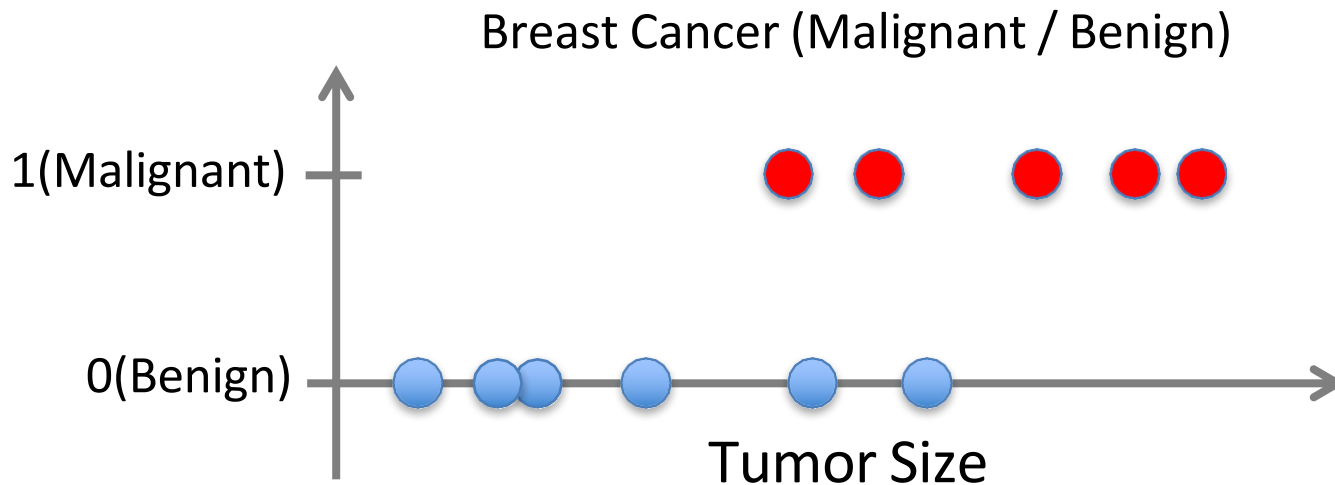
# Supervised Learning: Regression

- Given  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function  $f(x)$  to predict  $y$  given  $x$ 
  - $y$  is real-valued == regression



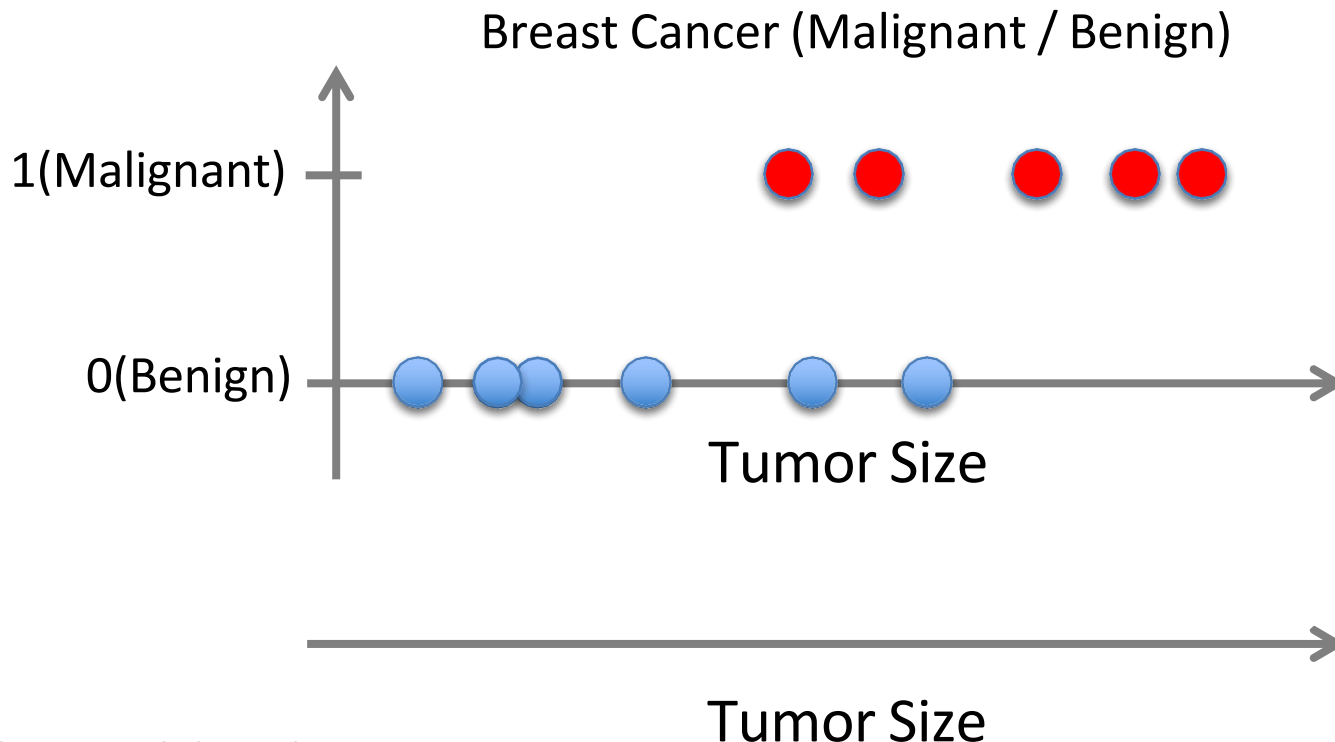
# Supervised Learning: Classification

- Given  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function  $f(x)$  to predict  $y$  given  $x$ 
  - $y$  is categorical == classification



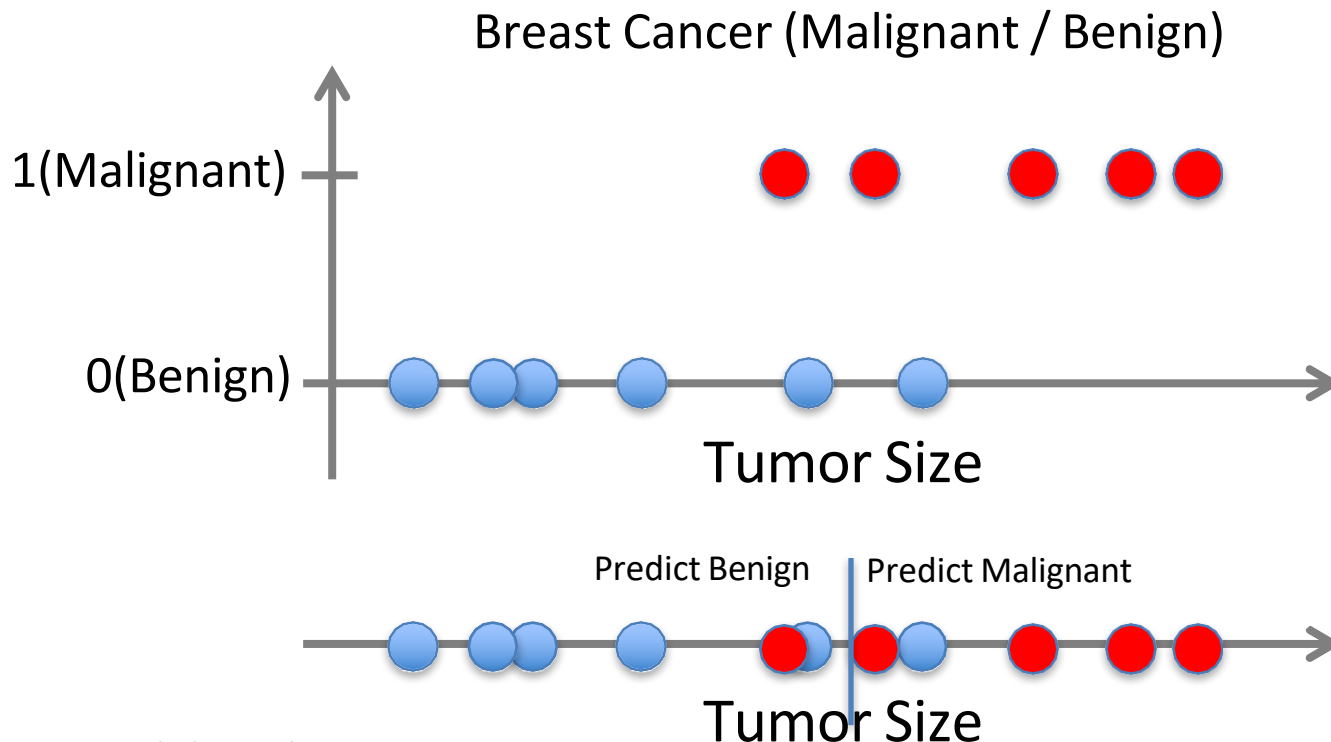
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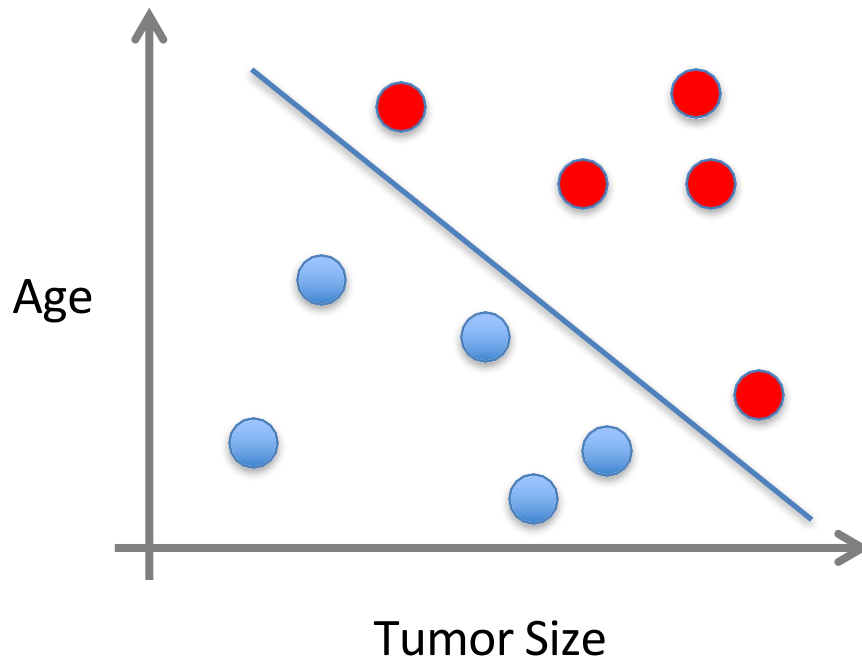
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# Supervised Learning

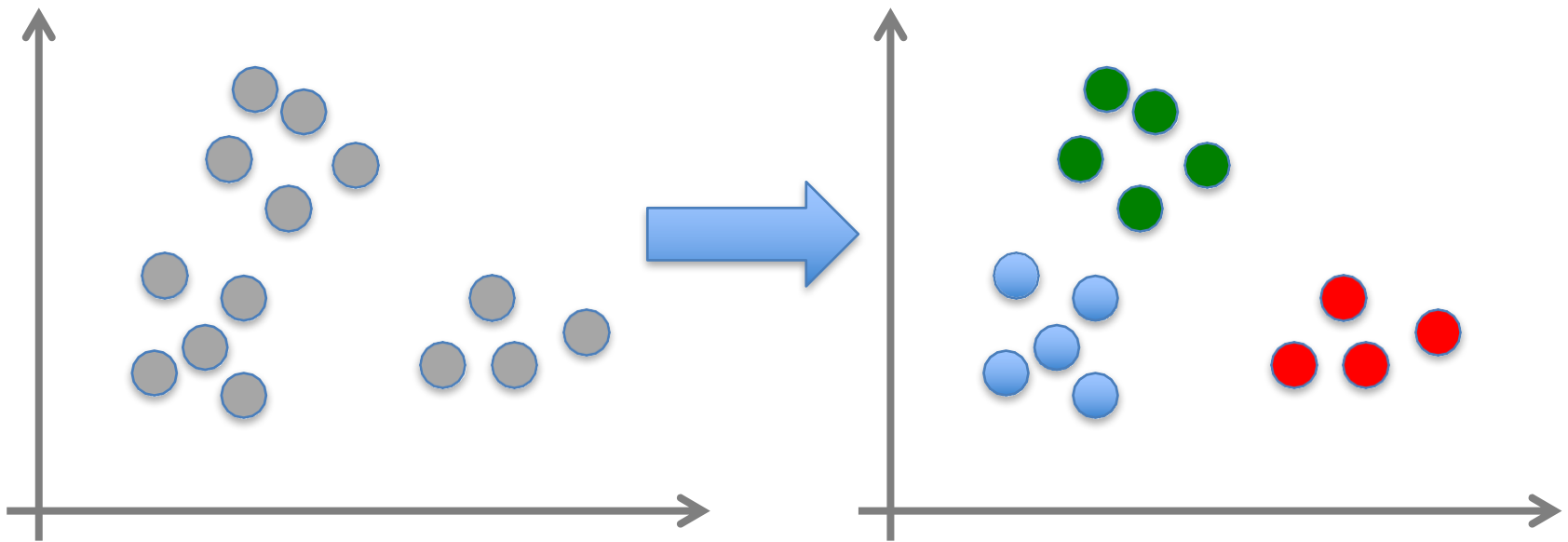
- $x$  can be multi-dimensional
  - Each dimension corresponds to an attribute



- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape
- ...

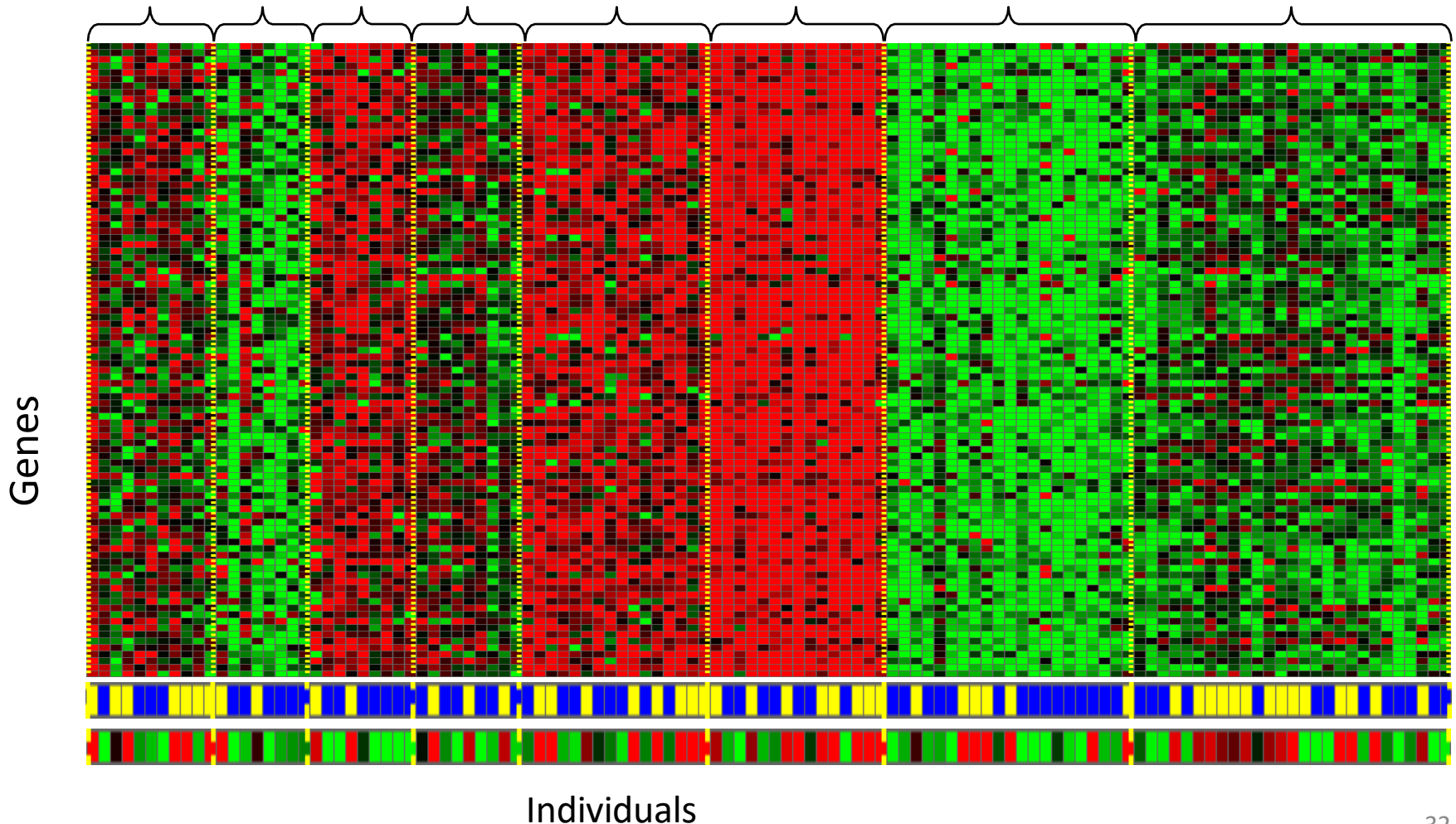
# Unsupervised Learning

- Given  $x_1, x_2, \dots, x_n$  (without labels)
- Output hidden structure behind the  $x$ 's
  - E.g., clustering



# Unsupervised Learning

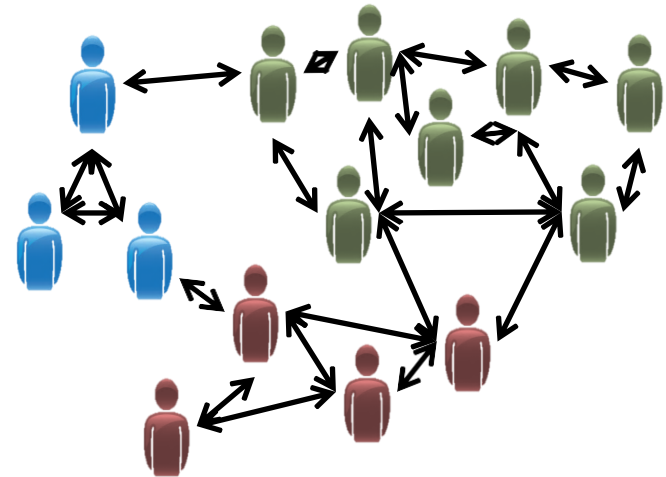
Genomics application: group individuals by genetic similarity



# Unsupervised Learning



Organize computing clusters



Social network analysis



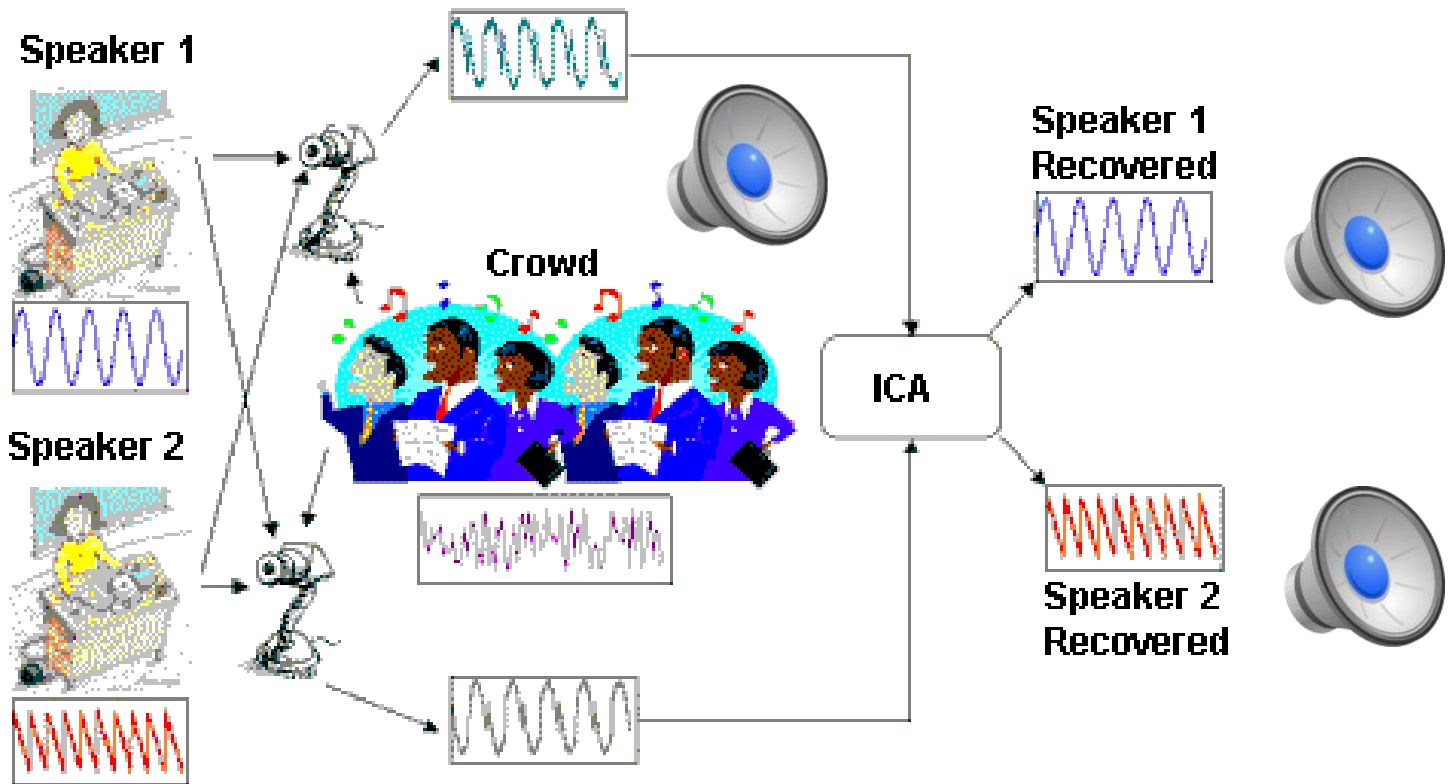
Market segmentation



Astronomical data analysis

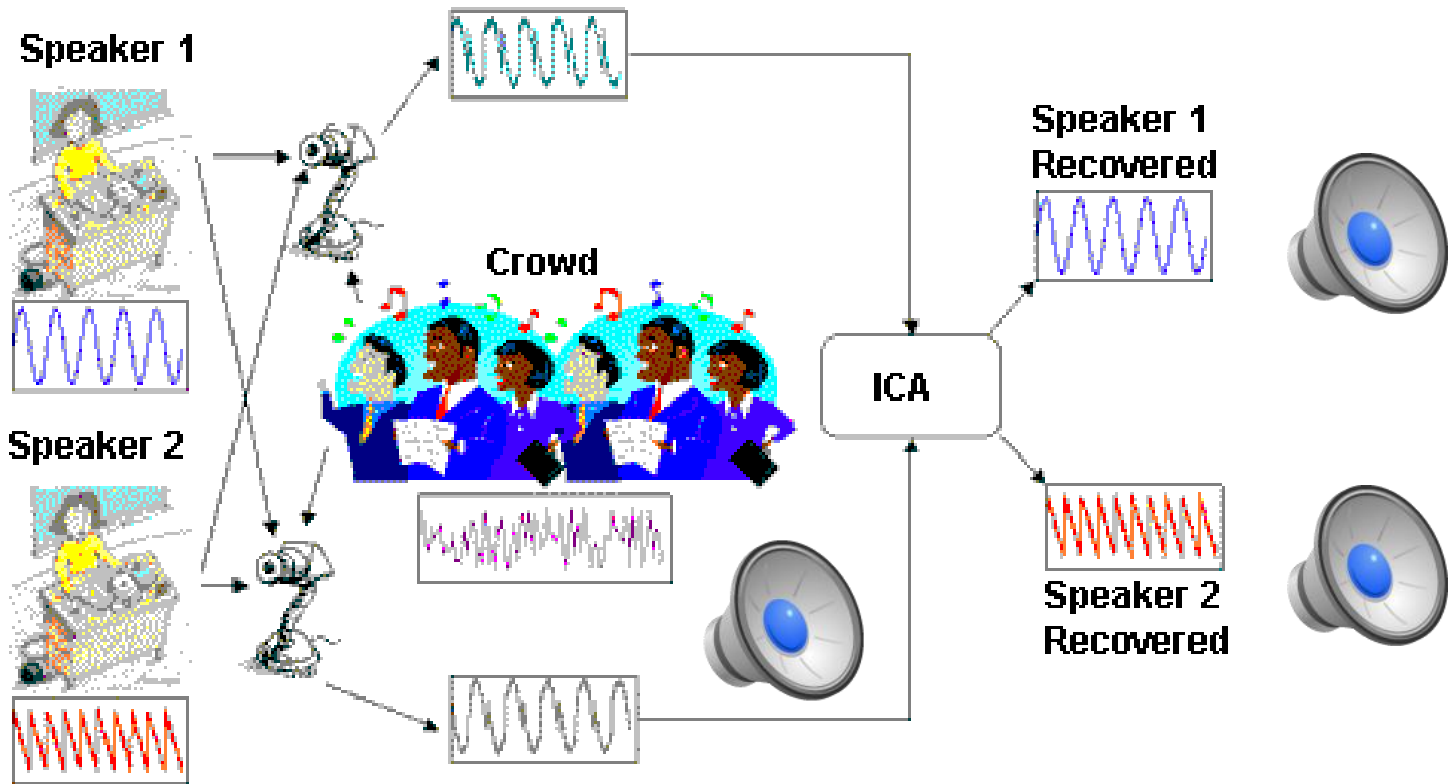
# Unsupervised Learning

- Independent component analysis – separate a combined signal into its original sources



# Unsupervised Learning

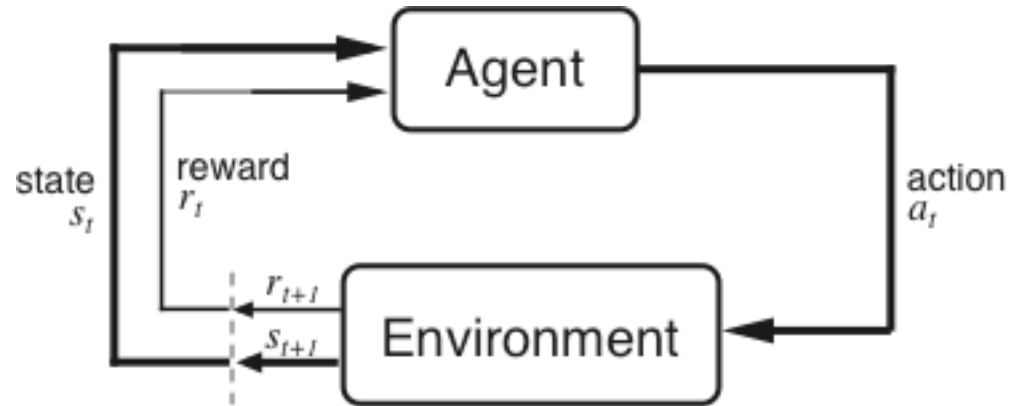
- Independent component analysis – separate a combined signal into its original sources



# Reinforcement Learning

- Given a sequence of states and actions with (delayed) rewards, output a policy
  - Policy is a mapping from states  $\rightarrow$  actions that tells you what to do in a given state
- Examples:
  - Credit assignment problem
  - Game playing
  - Robot in a maze
  - Balance a pole on your hand

# The Agent-Environment Interface



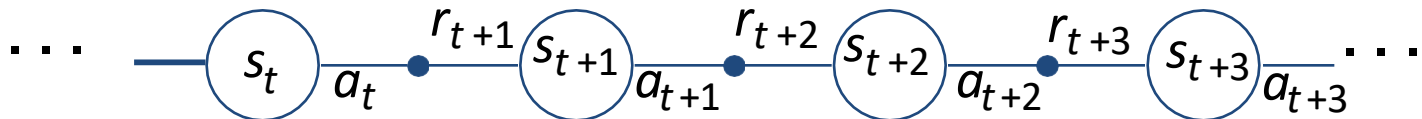
Agent and environment interact at discrete time steps :  $t = 0, 1, 2, K$

Agent observes state at step  $t$ :  $s_t \in S$

produces action at step  $t$ :  $a_t \in A(s_t)$

gets resulting reward :  $r_{t+1} \in \mathcal{R}$

and resulting next state :  $s_{t+1}$



# Reinforcement Learning



<https://www.youtube.com/watch?v=4cgWya-wjgY>

# Inverse Reinforcement Learning

- Learn policy from user demonstrations



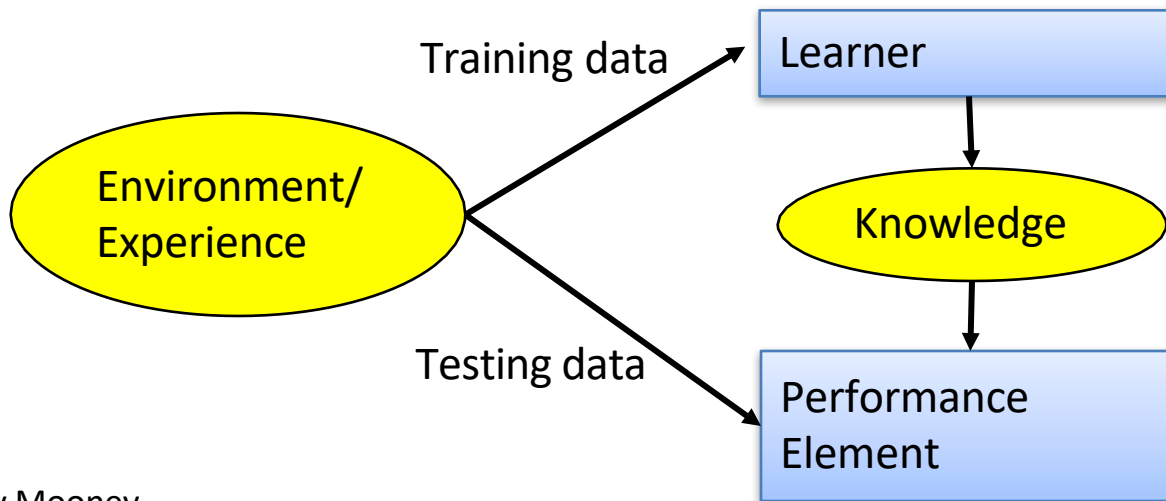
Fully Autonomous

<https://www.youtube.com/watch?v=GU16hXSSGKs>

# Framing a Learning Problem

# Designing a Learning System

- Choose the training experience
- Choose exactly what is to be learned
  - i.e. the *target function*
- Choose how to represent the target function
- Choose a learning algorithm to infer the target function from the experience



# Training vs. Test Distribution

- We generally assume that the training and test examples are independently drawn from the same overall distribution of data
  - We call this “i.i.d” which stands for “independent and identically distributed”
- If examples are not independent, requires *collective classification*
- If test distribution is different, requires *transfer learning*

# ML in a Nutshell

- Tens of thousands of machine learning algorithms
  - Hundreds new every year
- Every ML algorithm has three components:
  - **Representation**
  - **Optimization**
  - **Evaluation**

# Various Function Representations

- Numerical functions
  - Linear regression
  - Neural networks
  - Support vector machines
- Symbolic functions
  - Decision trees
  - Rules in propositional logic
  - Rules in first-order predicate logic
- Instance-based functions
  - Nearest-neighbor
  - Case-based
- Probabilistic Graphical Models
  - Naïve Bayes
  - Bayesian networks
  - Hidden-Markov Models (HMMs)
  - Probabilistic Context Free Grammars (PCFGs)
  - Markov networks

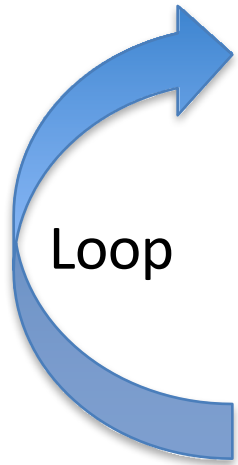
# Various Search/Optimization Algorithms

- Gradient descent
  - Perceptron
  - Backpropagation
- Dynamic Programming
  - HMM Learning
  - PCFG Learning
- Divide and Conquer
  - Decision tree induction
  - Rule learning
- Evolutionary Computation
  - Genetic Algorithms (GAs)
  - Genetic Programming (GP)
  - Neuro-evolution

# Evaluation

- Accuracy
- Precision and recall
- Squared error
- Likelihood
- Posterior probability
- Cost / Utility
- Margin
- Entropy
- K-L divergence
- etc.

# ML in Practice



- Understand domain, prior knowledge, and goals
- Data integration, selection, cleaning, pre-processing, etc.
- Learn models
- Interpret results
- Consolidate and deploy discovered knowledge

# Lessons Learned about Learning

- Learning can be viewed as using direct or indirect experience to approximate a chosen target function.
- Function approximation can be viewed as a search through a space of hypotheses (representations of functions) for one that best fits a set of training data.
- Different learning methods assume different hypothesis spaces (representation languages) and/or employ different search techniques.

# A Brief History of Machine Learning

# History of Machine Learning

- 1950s
  - Samuel's checker player
  - Selfridge's Pandemonium
- 1960s:
  - Neural networks: Perceptron
  - Pattern recognition
  - Learning in the limit theory
  - Minsky and Papert prove limitations of Perceptron
- 1970s:
  - Symbolic concept induction
  - Winston's arch learner
  - Expert systems and the knowledge acquisition bottleneck
  - Quinlan's ID3
  - Michalski's AQ and soybean diagnosis
  - Scientific discovery with BACON
  - Mathematical discovery with AM

# History of Machine Learning (cont.)

- 1980s:
  - Advanced decision tree and rule learning
  - Explanation-based Learning (EBL)
  - Learning and planning and problem solving
  - Utility problem
  - Analogy
  - Cognitive architectures
  - Resurgence of neural networks (connectionism, backpropagation)
  - Valiant's PAC Learning Theory
  - Focus on experimental methodology
- 1990s
  - Data mining
  - Adaptive software agents and web applications
  - Text learning
  - Reinforcement learning (RL)
  - Inductive Logic Programming (ILP)
  - Ensembles: Bagging, Boosting, and Stacking
  - Bayes Net learning

# History of Machine Learning (cont.)

- 2000s
  - Support vector machines & kernel methods
  - Graphical models
  - Statistical relational learning
  - Transfer learning
  - Sequence labeling
  - Collective classification and structured outputs
  - Computer Systems Applications (Compilers, Debugging, Graphics, Security)
  - E-mail management
  - Personalized assistants that learn
  - Learning in robotics and vision
- 2010s
  - Deep learning systems
  - Learning for big data
  - Bayesian methods
  - Multi-task & lifelong learning
  - Applications to vision, speech, social networks, learning to read, etc.
  - ???

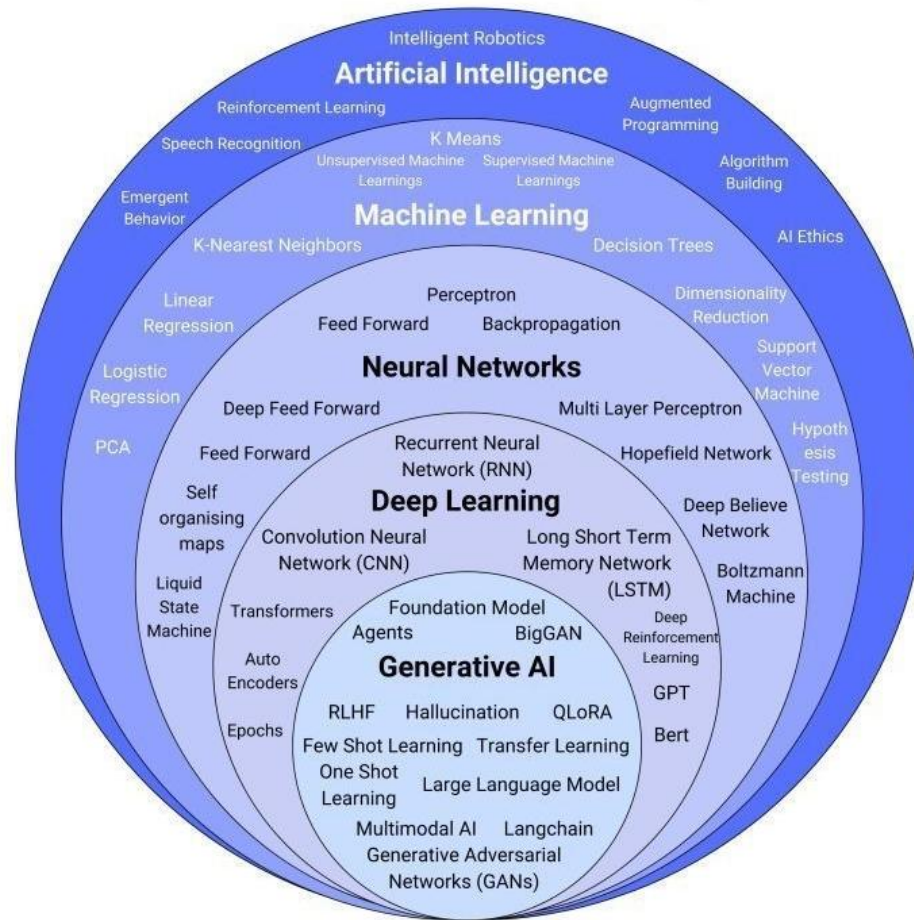
# What We'll Cover in this Course

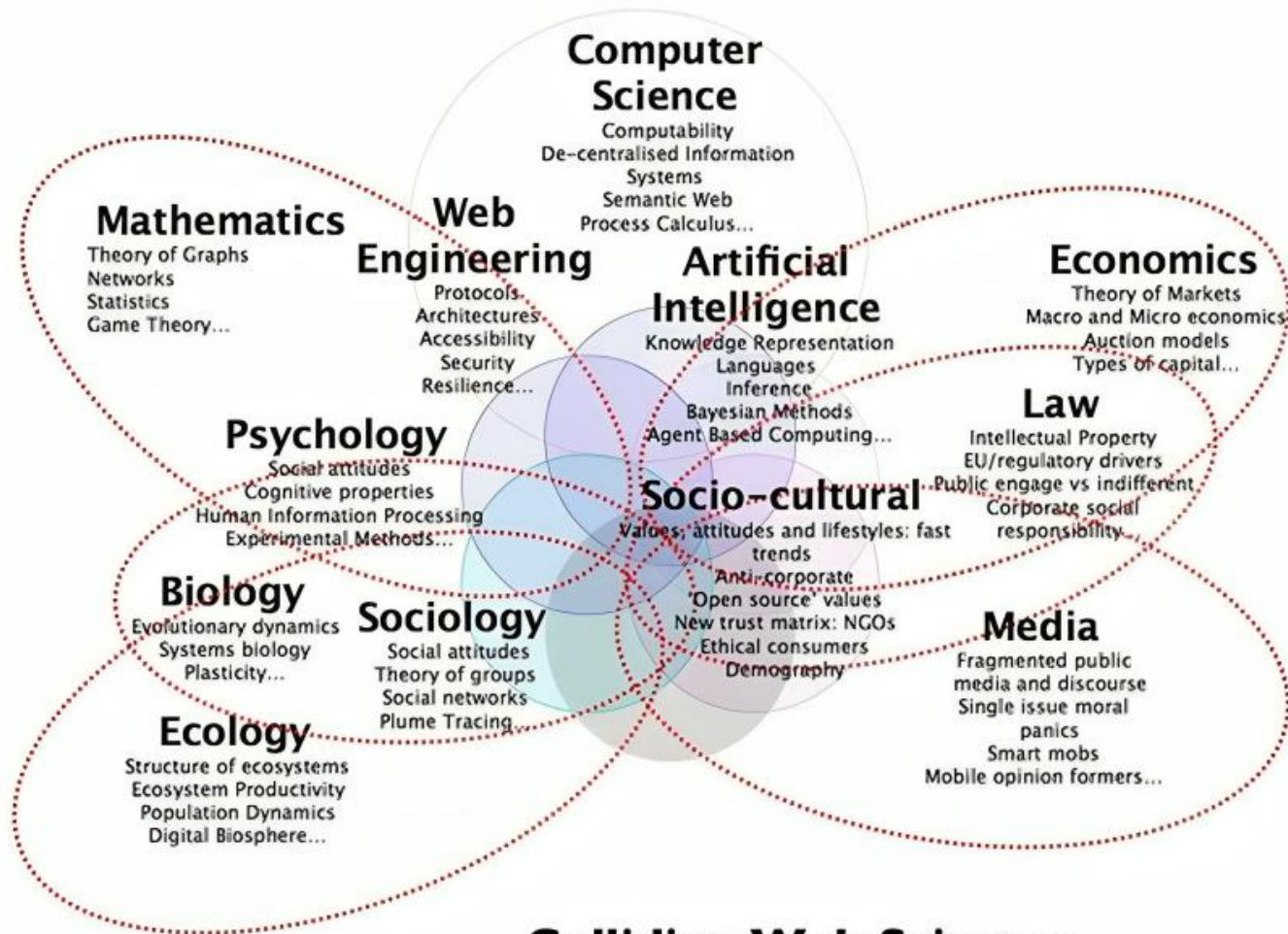
- **Supervised learning**
  - Decision tree induction
  - Linear regression
  - Logistic regression
  - Support vector machines & kernel methods
  - Model ensembles
  - Bayesian learning
  - Neural networks & deep learning
  - Learning theory
- **Unsupervised learning**
  - Clustering
  - Dimensionality reduction
- **Reinforcement learning**
  - Temporal difference learning
  - Q learning
- **Evaluation**
- **Applications**

Our focus will be on applying machine learning to real applications

|        |                                                                                                                                                                          |                                                                                                                                                                                                 |  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1950's |  <b>Artificial intelligence (AI)</b><br><i>Human intelligence exhibited by machines</i> |                                                                                                                                                                                                 |  |
|        | 1980's                                                                                                                                                                   |  <b>Machine learning</b><br><i>AI systems that learn from historical data</i>                                  |  |
|        | 2010's                                                                                                                                                                   |  <b>Deep learning</b><br><i>Machine learning models that mimic human brain function</i>                        |  |
|        | 2020's                                                                                                                                                                   |  <b>Generative AI (Gen AI)</b><br><i>Deep learning models (foundation models) that create original content</i> |  |

# The World of Artificial Intelligence





## Colliding Web Sciences

# Progress in Generative Models of Images – Diffusion Models



Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, 2021

# Ideogram, Dall-E, SORA

A photo of a Brazilian model with long, wavy hair and green eyes, aged 30. She is in the foreground, standing on a street with a yellow car parked beside her. The background reveals a blue building with white windows and a green tree. The pavement is made of cobblestones.





# 2 minutes Paper on GANs

Evolving Generative Adversarial Networks | Two Minute Papers #242



Two Minute Papers ✓  
1.61M subscribers

Join



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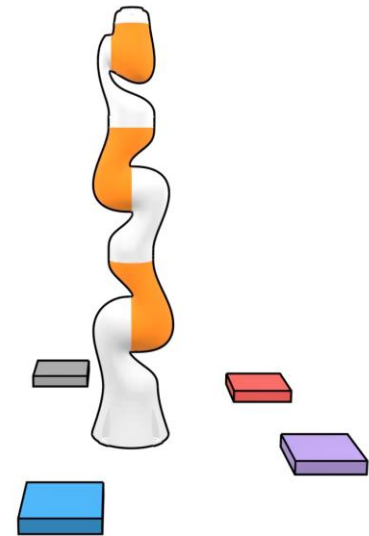


# Imitation Learning

Conditional generative model  $P(\text{actions} \mid \text{past observations})$

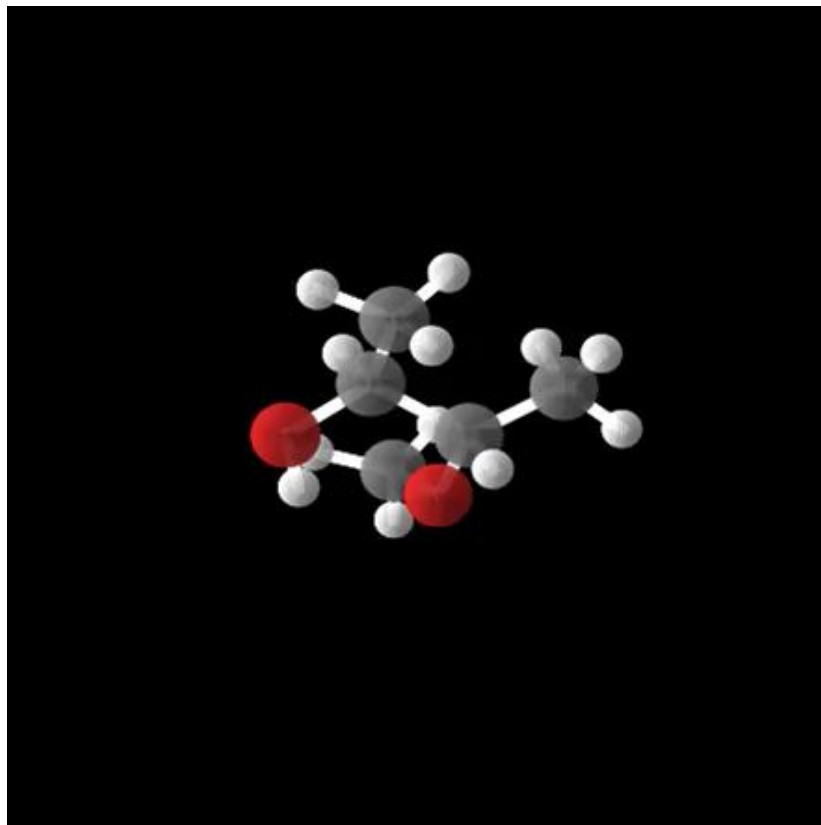


Li et al., 2017



Janner et al., 2022

# Molecule generation



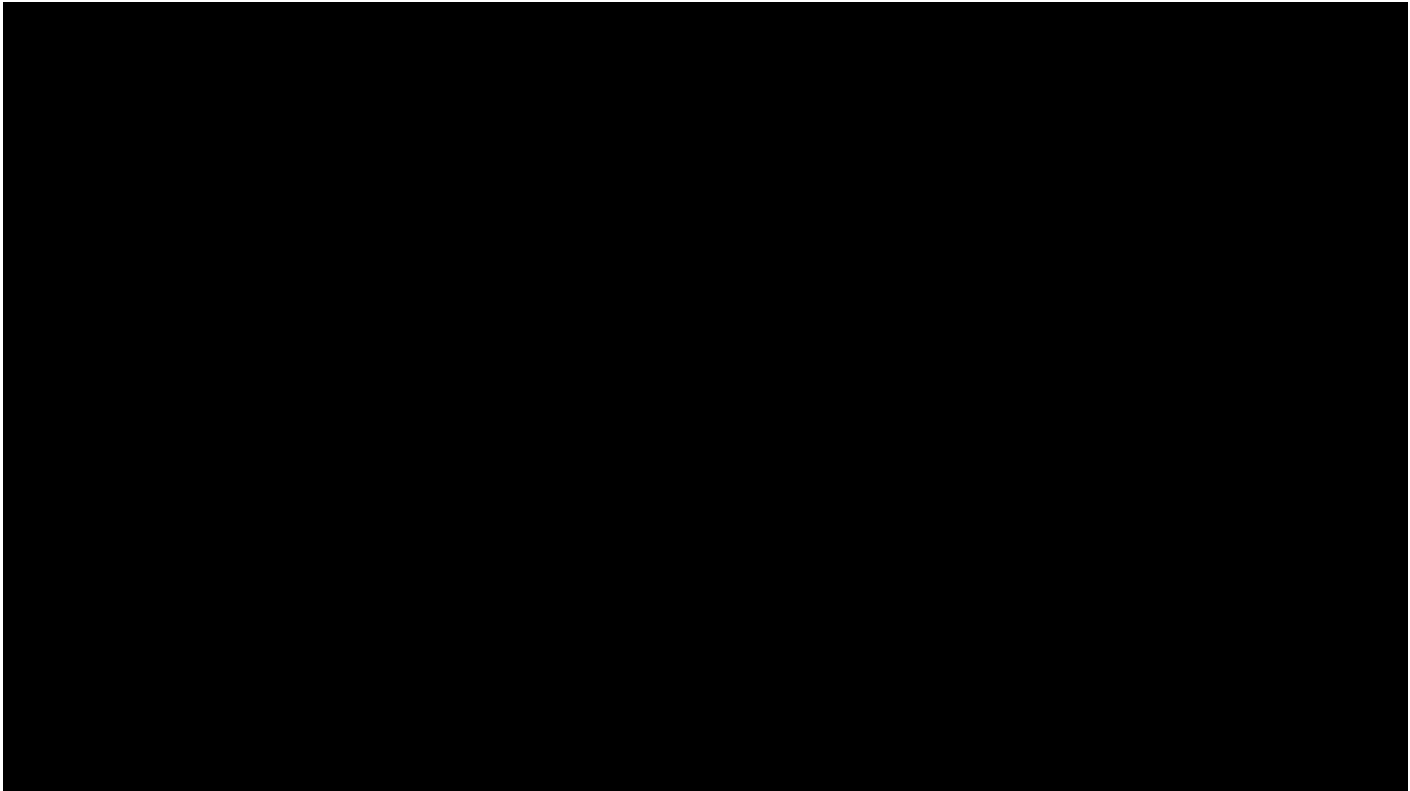
# DeepFakes

Which image is real?



@QaziMazhar\_

# DeepFakes



# Projects



# Course Policy

- Participation
- In-class attendance/ participation
- Plagiarism
  - Do NOT do it!!!
- AI writing detection should be below 20% in final report.



# Projects

- Course projects will be done in groups of up to 3 max students and can fall into one or more of the following categories:
- Search one recent paper that uses Machine learning method in your research area and bring it next class or real world problem solution through ML or AI.
- Each class we will have discussion on Projects and each group discuss one page of progress every week.
- Each group should be able to complete writing at least first draft before 14 week.





## Projects

[https://docs.google.com/spreadsheets/d/1SZY0186L\\_Fuv5fxdMn2XlrDZG\\_xlJPUnhZvFjkYYajQ/edit?usp=sharing](https://docs.google.com/spreadsheets/d/1SZY0186L_Fuv5fxdMn2XlrDZG_xlJPUnhZvFjkYYajQ/edit?usp=sharing)



To access the project excel Sheet



# Create your first chatbot

Link to Google Colab;

<https://colab.research.google.com/drive/1XlBqkWLxOkGn9MiUVc2bL88XN-7EeBeh#scrollTo=8M4oDotXQW1z>

Login with your Gmail not institutional email